

Algebra: Finding the Inverse of a Function

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DIRECTIONS

Find $f^{-1}(x)$: swap x and y , then solve for y . Simplify completely.

- 1 Find the inverse function $f^{-1}(x)$:

$$f(x) = \frac{x+3}{5}$$

Answer: _____

- 2 Find the inverse function $f^{-1}(x)$:

$$f(x) = 5x + 1$$

Answer: _____

- 3 Find the inverse function $f^{-1}(x)$:

$$f(x) = x^3$$

Answer: _____

- 4 Find the inverse function $f^{-1}(x)$:

$$f(x) = 2x + 6$$

Answer: _____

- 5 Find the inverse function $f^{-1}(x)$:

$$f(x) = 3x - 9$$

Answer: _____

- 6 Find the inverse function $f^{-1}(x)$:

$$f(x) = 2x - 10$$

Answer: _____

- 7 Find the inverse function $f^{-1}(x)$:

$$f(x) = \frac{x}{4} + 2$$

Answer: _____

- 8 Find the inverse function $f^{-1}(x)$:

$$f(x) = \sqrt{x-1}$$

Answer: _____

- 9 Find the inverse function $f^{-1}(x)$:

$$f(x) = \frac{2}{x}$$

Answer: _____

- 10 Find the inverse function $f^{-1}(x)$:

$$f(x) = 4x - 7$$

Answer: _____



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Answer Key & Solutions

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TEACHER NOTES Verify with $(f \circ f^{-1})(x) = x$. Restrict domain for non-one-to-one functions like square roots.

1 Find the inverse function $f^{-1}(x)$:

$$= f^{-1}(x) = 5x - 3$$

$$f(x) = \frac{x+3}{5}$$

2 Find the inverse function $f^{-1}(x)$:

$$= f^{-1}(x) = \frac{x-1}{5}$$

$$f(x) = 5x + 1$$

3 Find the inverse function $f^{-1}(x)$:

$$= f^{-1}(x) = \sqrt[3]{x}$$

$$f(x) = x^3$$

4 Find the inverse function $f^{-1}(x)$:

$$= f^{-1}(x) = \frac{x-6}{2}$$

$$f(x) = 2x + 6$$

5 Find the inverse function $f^{-1}(x)$:

$$= f^{-1}(x) = \frac{x+9}{3}$$

$$f(x) = 3x - 9$$

6 Find the inverse function $f^{-1}(x)$:

$$= f^{-1}(x) = \frac{x+10}{2}$$

$$f(x) = 2x - 10$$

7 Find the inverse function $f^{-1}(x)$:

$$= f^{-1}(x) = 4(x - 2)$$

$$f(x) = \frac{x}{4} + 2$$

8 Find the inverse function $f^{-1}(x)$:

$$= f^{-1}(x) = x^2 + 1, x \geq 0$$

$$f(x) = \sqrt{x-1}$$

9 Find the inverse function $f^{-1}(x)$:

$$= f^{-1}(x) = \frac{2}{x}$$

$$f(x) = \frac{2}{x}$$

10 Find the inverse function $f^{-1}(x)$:

$$= f^{-1}(x) = \frac{x+7}{4}$$

$$f(x) = 4x - 7$$

